RESEARCH PROJECTS

Research Structure & Scope

Obj 3: Data-centric AI

- Enhance data quality & transparency

Obj 4: Privacy/Security

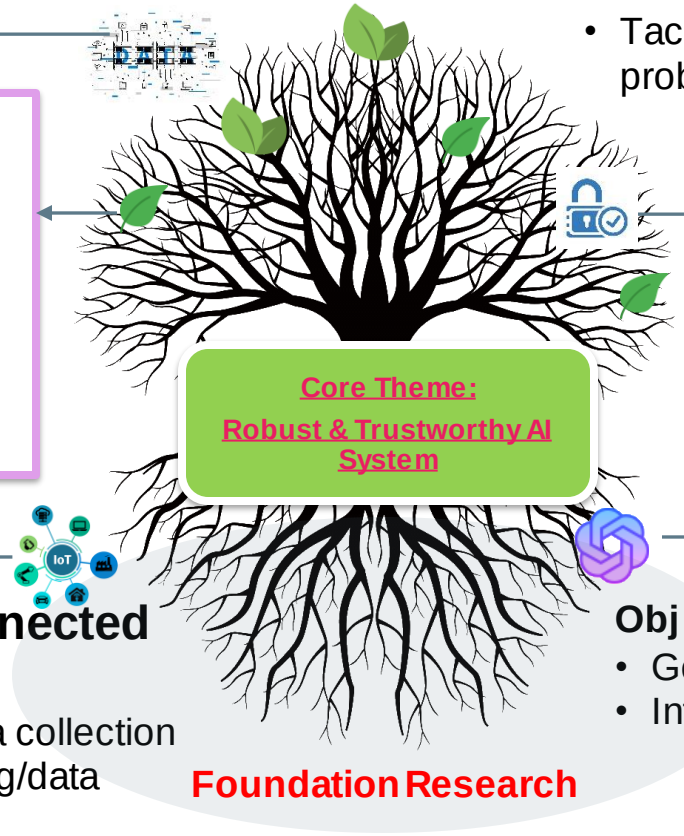
Federated Learning

- Decentralize privacy and security in model building
- Tackle adversarial attacks & Non-IID problem

Branch Research

Future research direction:

- Quantifying and enhancing trustworthiness
- Robustness against adversarial attacks
- Human-AI collaboration and explainability



Obj 1: IoT-based Connected System

- Enable heterogeneous data collection
- Provide real-time monitoring/data analysis
- Smart Edge device efficiency

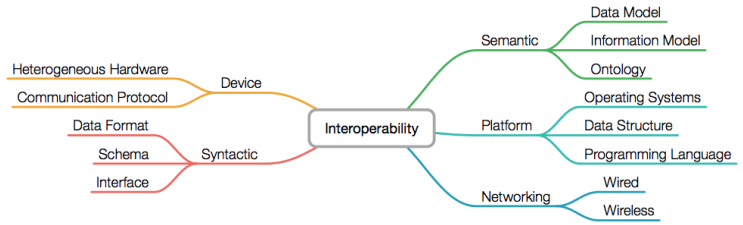
Obj 2: Robust Generative AI

- Generate diverse/reliable/trustworthy data
- Integrating context conditions

Foundation Research

OBJECTIVE 1: SMART IOT SYSTEM FOR DATA COLLECTION & MONITORING

- **1. Interoperability issue** in IoT systems
- **2. Offline/Intermittent/non-scalable data analysis** for care monitoring



Interoperability issues in IoT.

Real-time at home monitoring

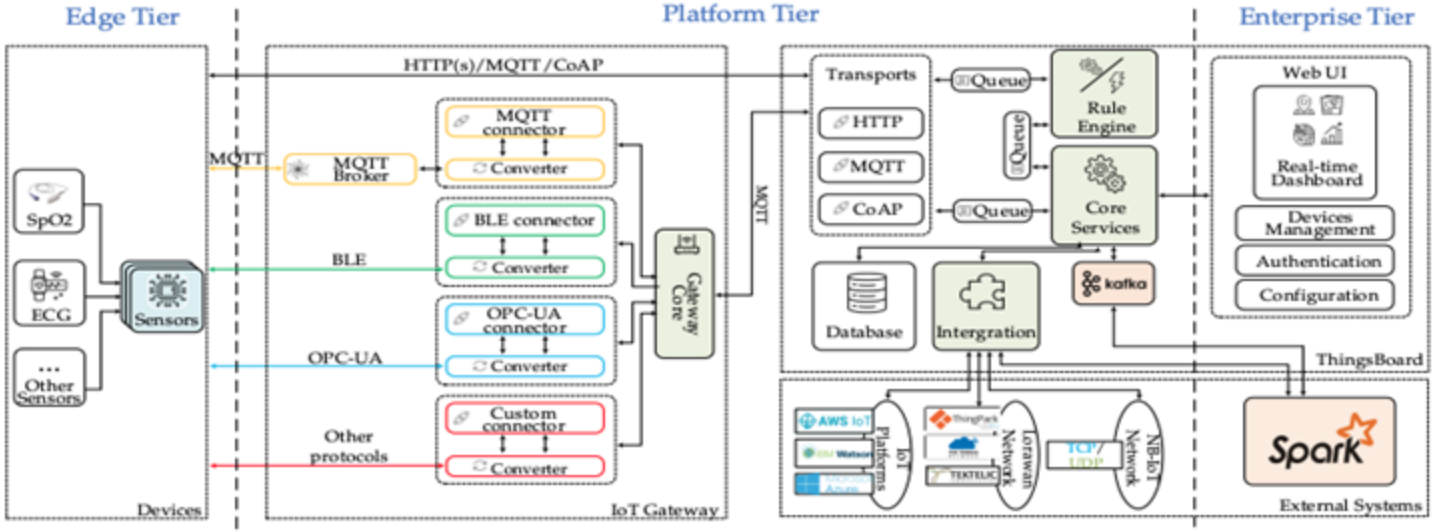
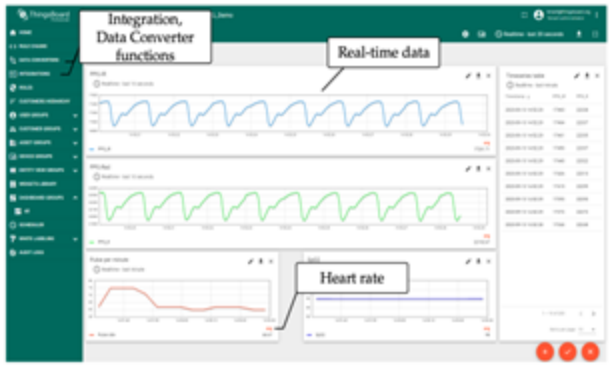


Fig. 4. The proposed system architecture based on IIRA three-tier model



(a) (b) (c)

Figure 4: Actual PCB enclosure design. Subfigure (a) shows the SpO2 device with enclosure, (b) depicts the SpO2 device with a human subject, and (c) displays a setup of the SpO2 device with a simulator.

OBJECTIVE 2: CONDITIONAL DIFFUSION MODEL-BASED MRI IMAGING FRAMEWORK

How to **Generate realistic and contextually relevant brain tumor images** in medical image synthesis ?

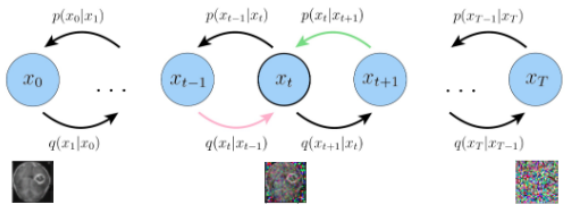
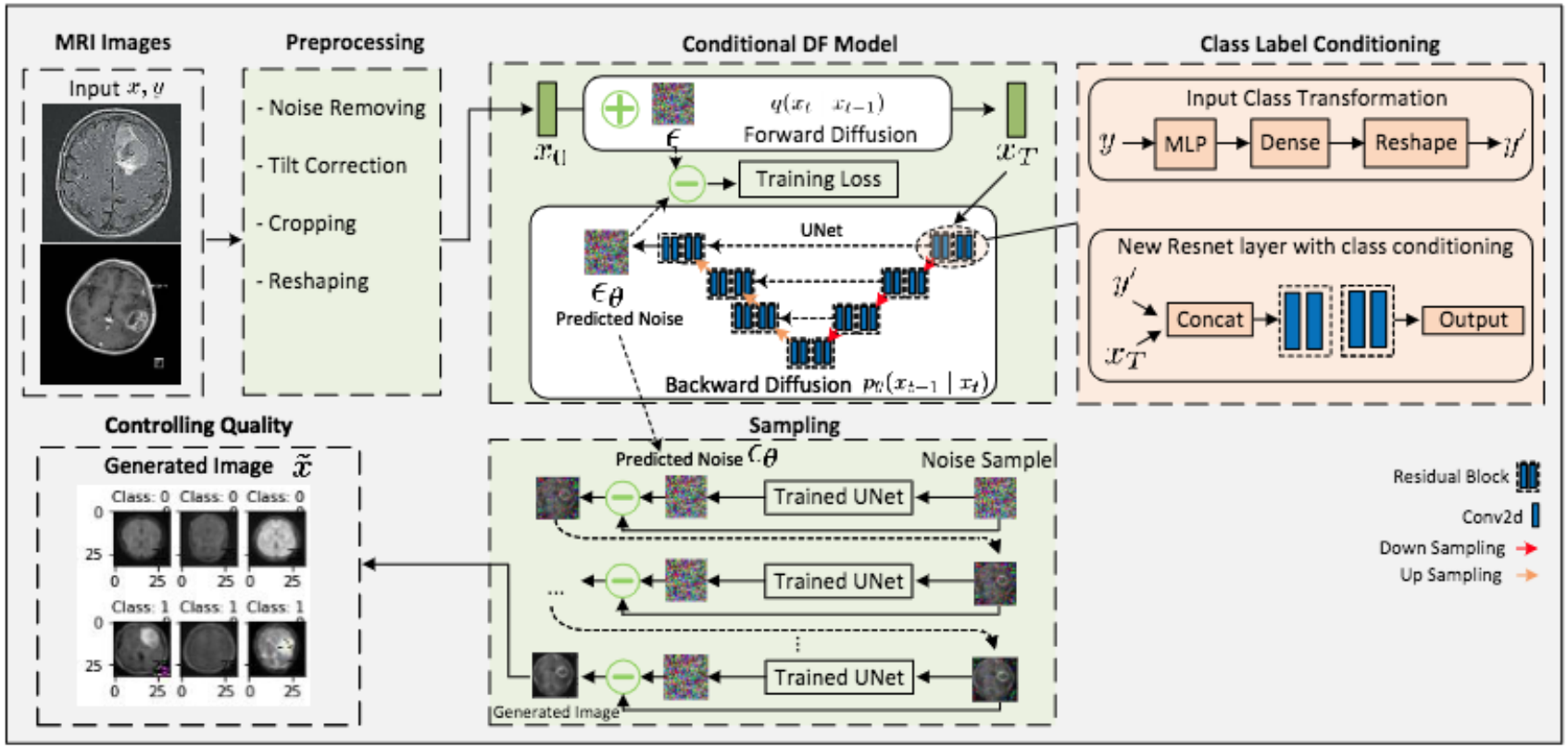


Fig. 1: Visual representation of a Diffusion Model. An input is steadily noised over time until it becomes identical to Gaussian noise; a diffusion model learns to reverse this process. In this figure, for each intermediate latent, we minimize the difference between the distributions represented by the pink and green arrows.



OBJECTIVE 3. HYBRID DATA-CENTRIC AND MODEL-CENTRIC FOR AI DEVELOPMENT

This work presents a hybrid data-centric and model-centric system for AI development. The core idea is to build a robustness AI system, not only model-centric, but also data-centric should be taken into account. We proposed data-centric, and model centric methods to improve and tune input data, as well as model quality. OSA detection for cancer patients is a case-study to demonstrate for the effectiveness of the proposed system. The detection algorithm is based on stacked LSTM, which is considered as a cross-silo model for Federated Learning.

2. **Huu-Quoc Nguyen**, Trung Tim Le, "Hybrid Data-centric and Model-centric for Robustness Single-Lead ECG Obstructive Sleep Apnea Diagnosis with Federated Learning", IEEE Journal of Biomedical and Health Informatics, 2022 (On-going)

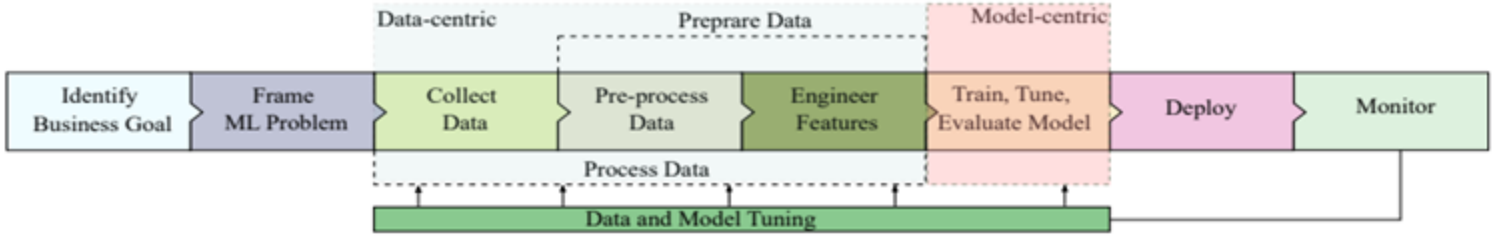
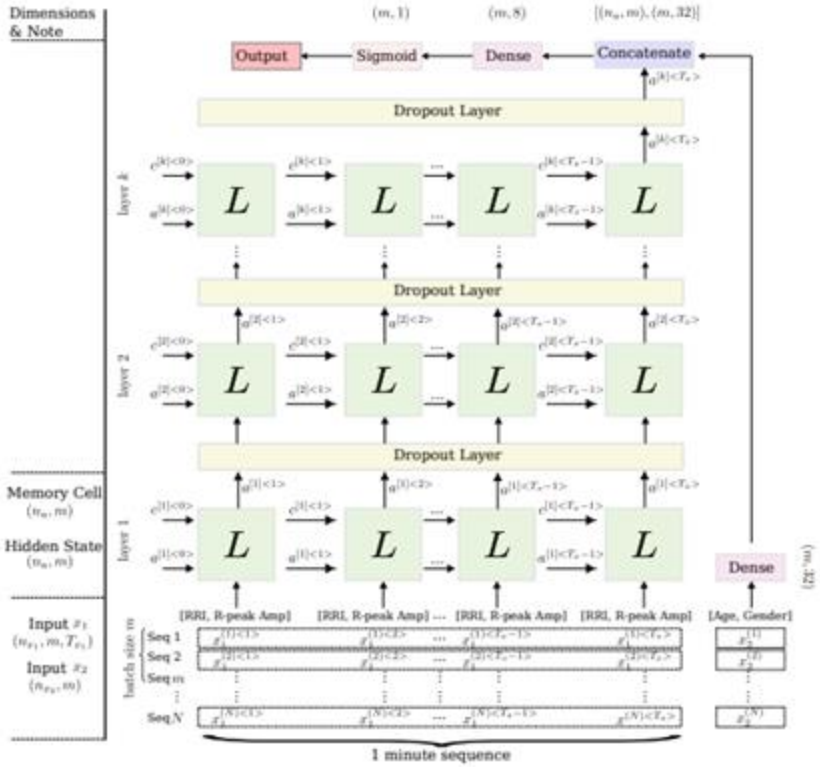


Fig. 1: A robustness ML life-cycle phases with hybrid-centric approaches.



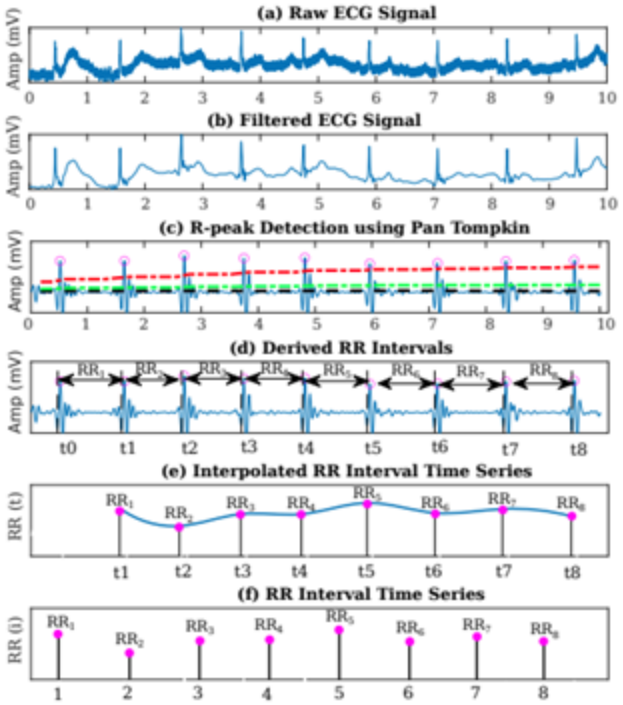
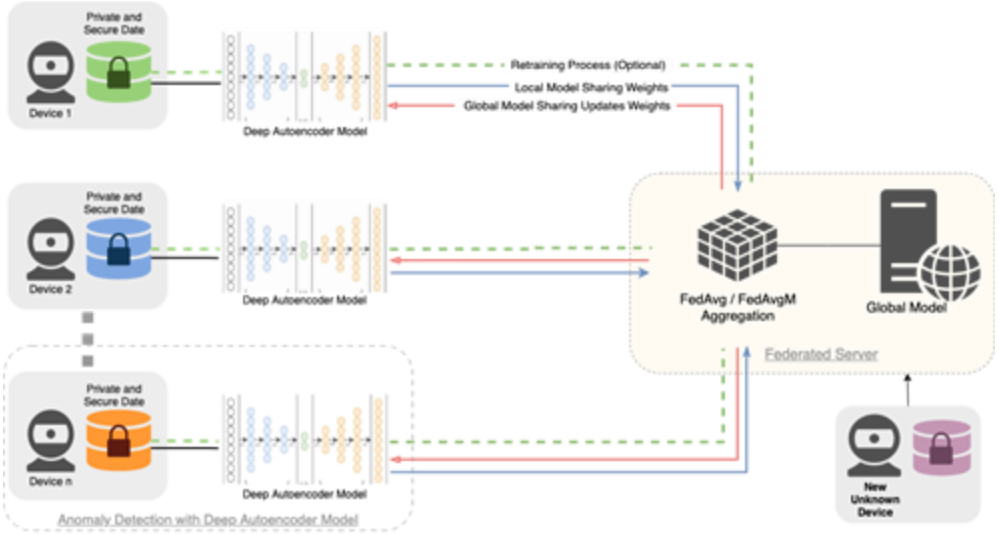
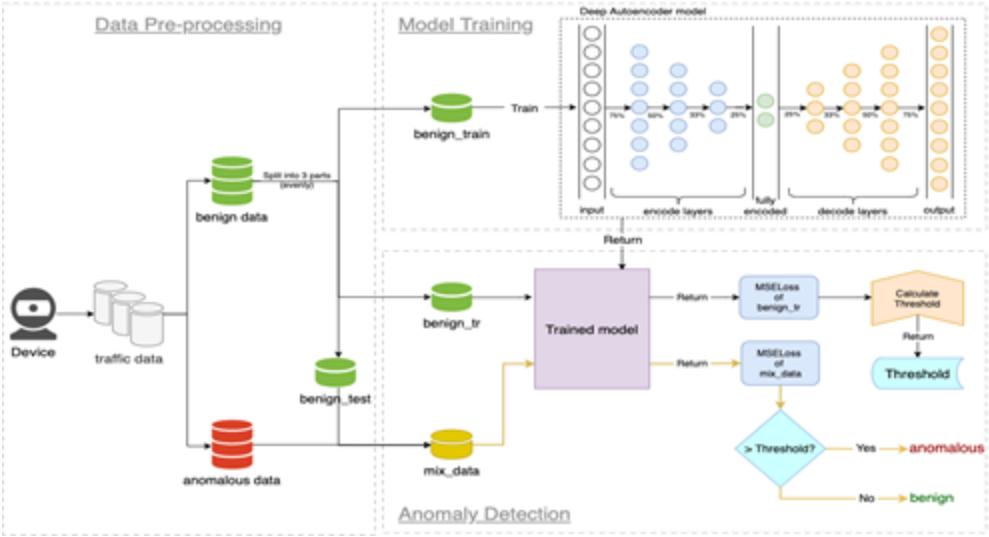


Fig. 4: The proposed stacked LSTM architecture for OSA detection. The network consisted of ....The input x1, x2 ...., the LSTM layers and the output ....

OBJECTIVE 4. PRIVACY FEDERATED LEARNING FOR INTERNET OF THINGS

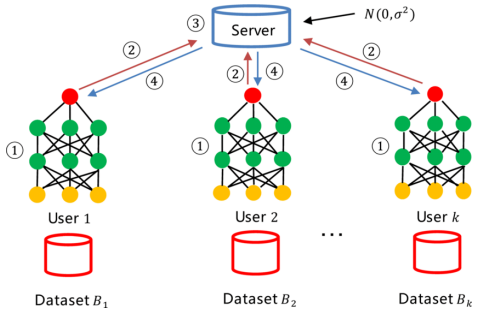
Traditional, centralized machine-learning models have a multitude of limitations. The need to share data from remote clients to a main central server causes high latency and lengthy training times. Furthermore, the sharing of data poses a security risk. This is particularly worrisome for models including Internet of Medical Things devices, which contain sensitive data. Federated learning is a decentralized type of machine learning, where the learning mainly occurs at each client. A learning parameter is then sent from each client to a central server, where the local models are aggregated. This allows for improved training times and data security, by decreasing the risk of data interception. This project utilizes the N-BaIoT dataset, which features nine different household IoT devices undergoing both Mirai and BASHLITE types of attacks. The anomalies are detected using an autoencoder algorithm. The data is preprocessed using min-max normalization for each feature and a threshold is then developed using the reconstruction mean square error of the autoencoder. If the reconstruction error is significant, the data's equation surpasses the threshold and is classified as anomalous.

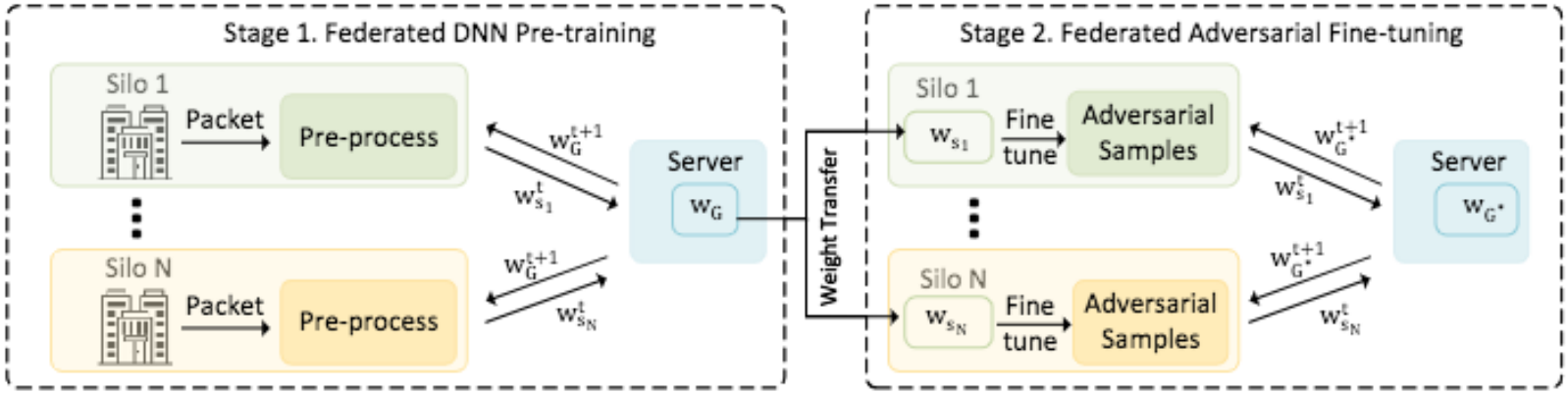




OBJECTIVE 4: FEDNIDS – FEDERATED LEARNING FRAMEWORK FOR NIDS

How to build a robust NIDS based on packet data?
This study aims to address this limitation by **focusing on packet-based data**, allowing for the detection of more intricate and transient attack behaviors in real-time. By **leveraging** Deep Neural Networks (**DNNs**) and Federated Learning (**FL**), the goal is to **enhance the accuracy, robustness, and privacy of packet-based NIDS.**





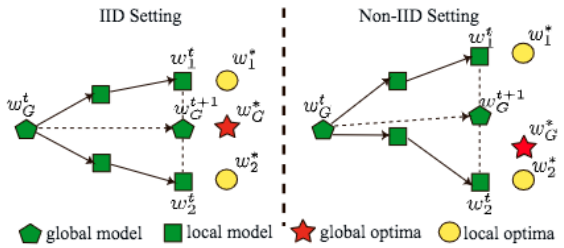


Fig. 1. Example of a drift under the non-IID setting

Table 6. Performance comparison of FedNIDS with other approaches

Silo	Class	FedAvg			Centralized Approach			FedNIDS		
		Precision	Recall	F1 Score	Precision	Recall	F1 Score	Precision	Recall	F1 Score
Silo 1	Attack	1.00	0.97	0.98	0.99	0.98	0.99	**1.00**	**0.99**	**0.99**
	Benign	0.95	1.00	0.98	0.97	0.99	0.98	1.00	1.00	1.00
Silo 2	Attack	1.00	0.96	0.98	0.99	1.00	0.99	**1.00**	**0.99**	**0.99**
	Benign	0.95	1.00	0.97	0.99	1.00	0.99	0.96	0.93	0.95
Silo 3	Attack	1.00	0.68	0.80	0.99	0.90	0.95	**1.00**	**0.95**	**0.96**
	Benign	0.75	1.00	0.86	0.91	0.99	0.95	0.97	1.00	0.96
Silo 4	Attack	1.00	0.67	0.80	0.99	0.90	0.94	**1.00**	**0.96**	**0.94**
	Benign	0.75	1.00	0.86	0.90	0.99	0.94	0.98	0.97	0.95

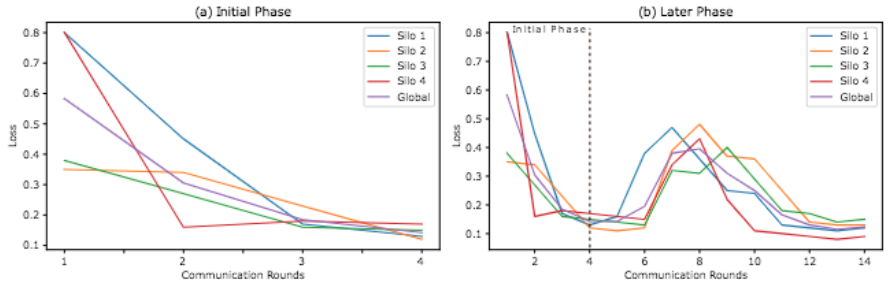
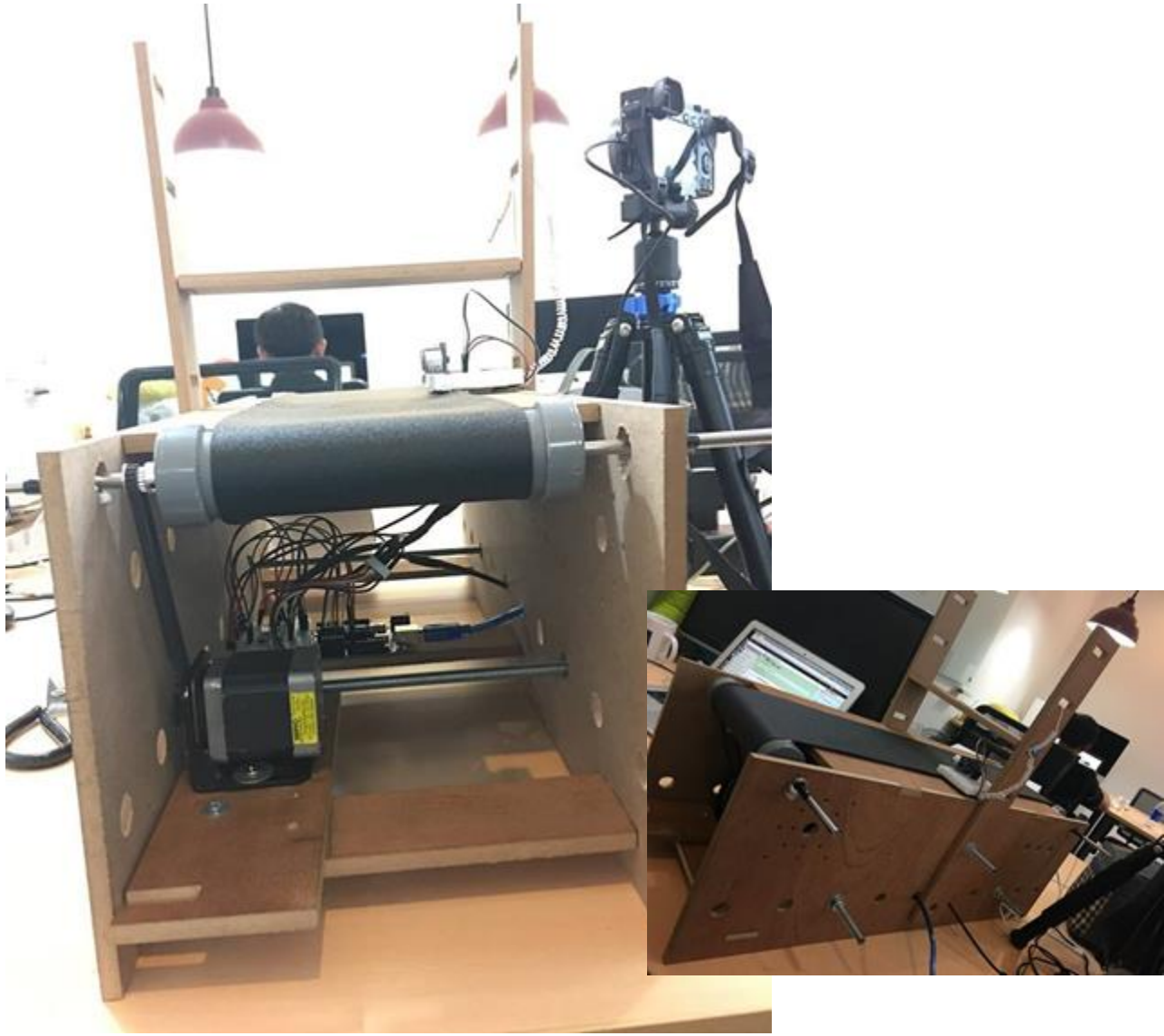


Fig. 3. FedNIDS adaptation to emerging threats.

INDUSTRY PROJECTS

1. CONVEYOR BELT
(IoT, C++, Machine Learning | PROJECT LEADER)

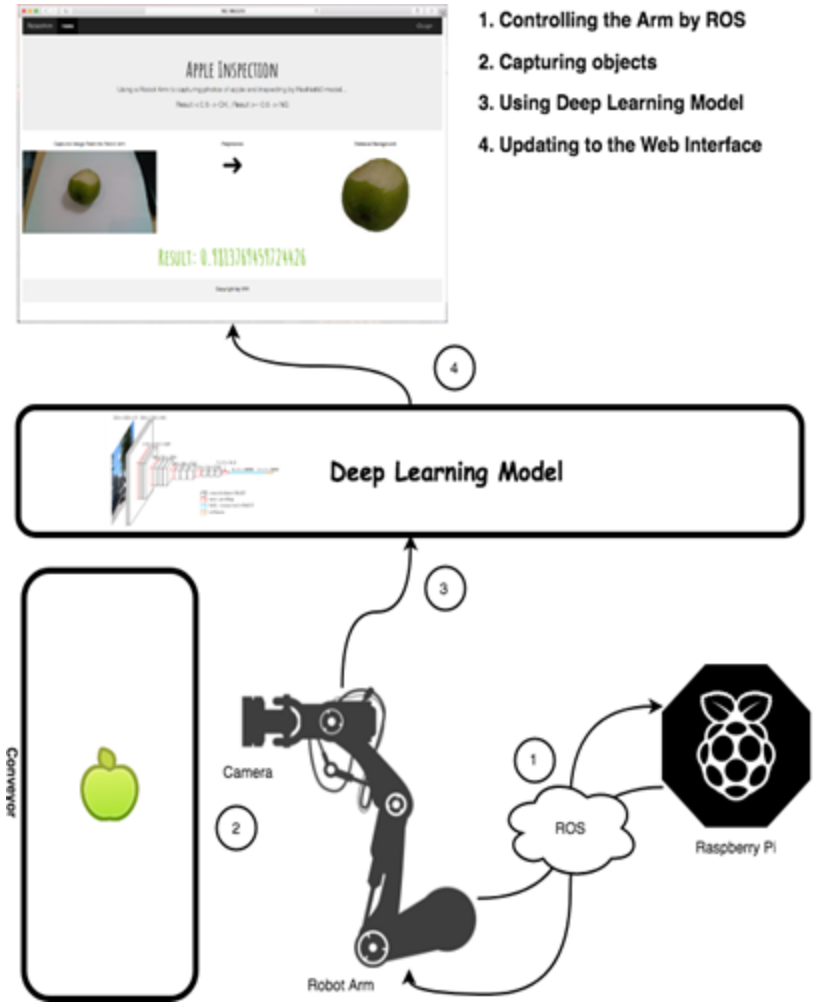
As what we know, Machine Learning depends heavily on data. But taking photos for data collection consume a lot of time. Sometimes it takes months before the first algorithm is built.
-> Conveyor Belt project aims making the data-preparation becomes more automatic and efficient, as well as cost savings of manual collecting data.
Video:
https://drive.google.com/open?id=1SseCgRfcX7_vHNinsQlDrk-jccfvDHLC

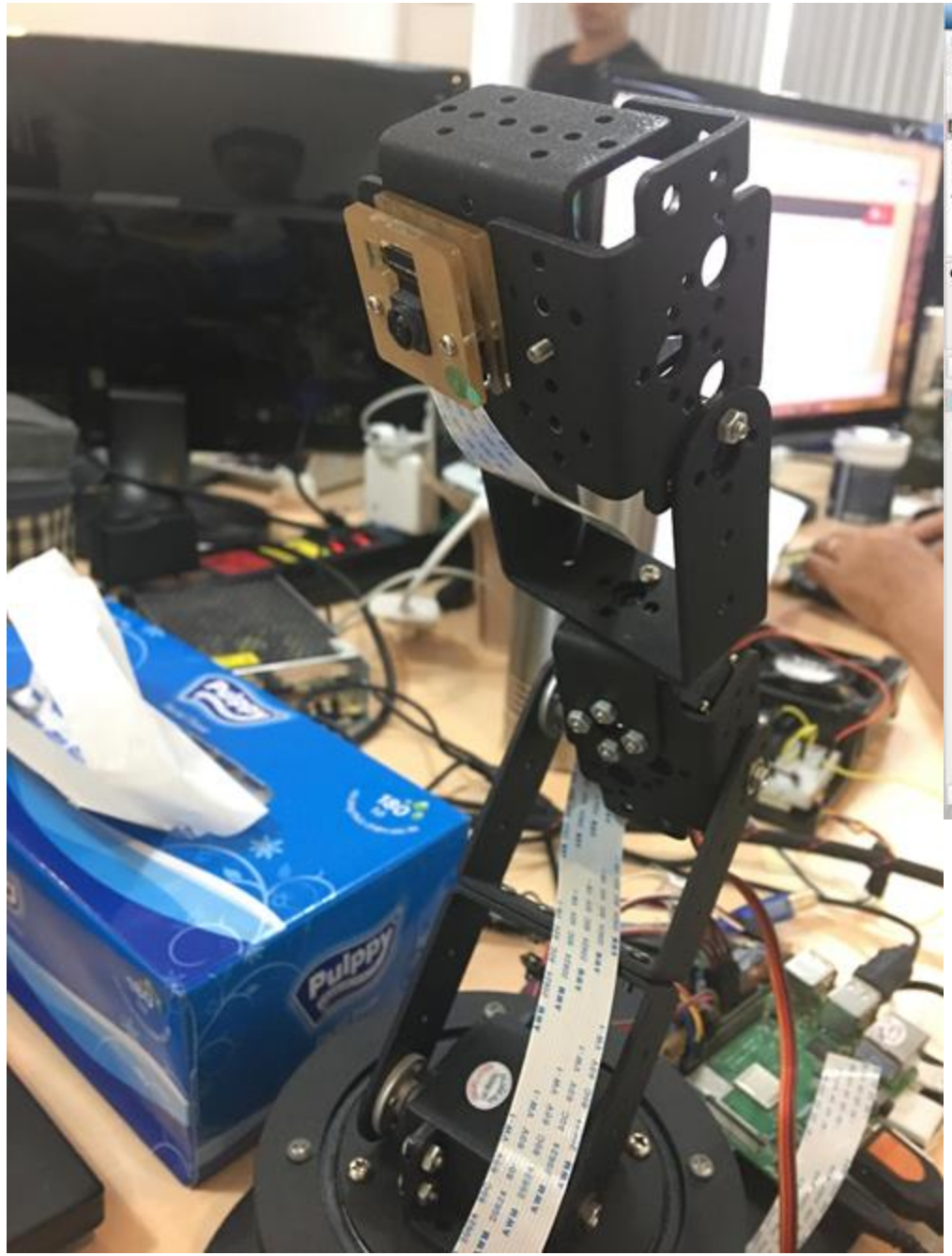


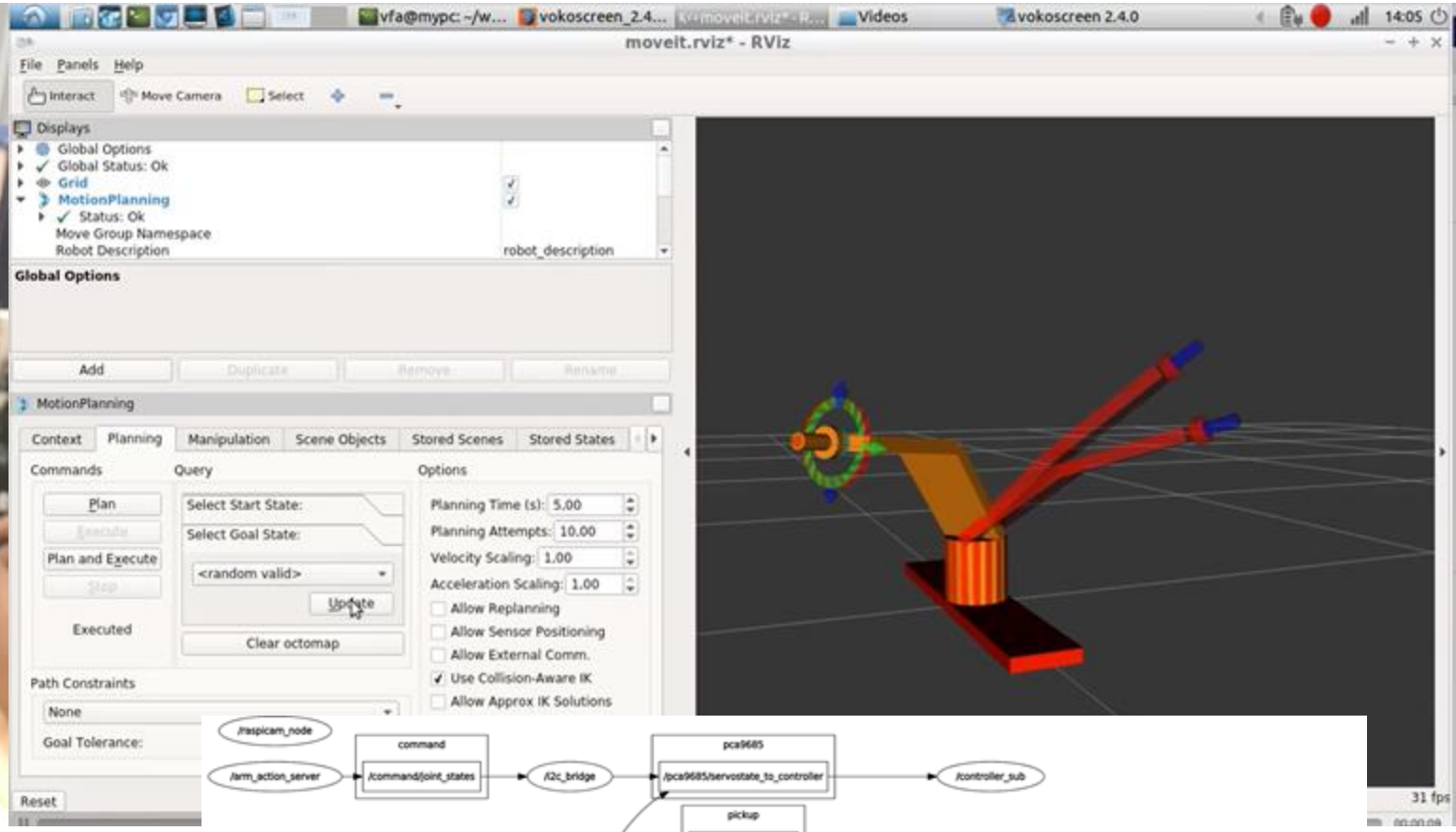
2. ROBOT ARM
(IoT, Deep Learning, ROS, Machine Learning)

In factories, product is fed onto the conveyor belt where grading personnel stand along each side and visually inspect the product for defects. Defect product is manually removed and dropped into the trash lanes along the edge of each side of the conveyor. Good product continues along the length of the conveyor and discharges off of the end of the belt while defects are dropped through a trash chute at the exit end of the conveyor and into a lug. But the problem is this kind of system needs a lot of skill-workers to manually remove defects, so it should be automated from A to Z. In this situation, building a Robot Arm for checking defects is essential and promising. A Robot Arm does in several hours what it would take labor 2-3 days to accomplish.

Source:
https://drive.google.com/file/d/1sIMJ3BOCjd-B2ghHndZeDxoD2xsCqL9v/view



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Status: Ok
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Rename
MotionPlanning
Context
Planning
Manipulation
Scene Objects
Stored Scenes
Stored States
Commands
Query
Options
Plan
Execute
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Stop
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Select Start State:
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Planning Attempts: 10.00
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Acceleration Scaling: 1.00
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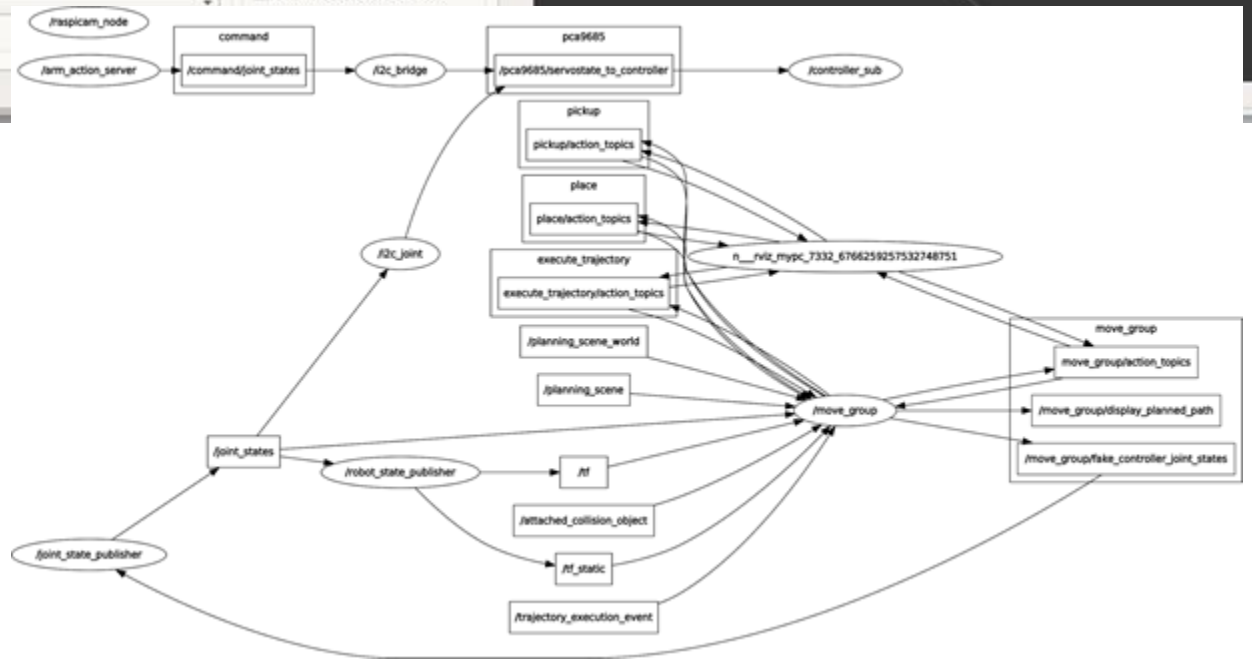
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3. CHANGING HAIR COLOR
(Deep Learning, iOS, Python | PROJECT LEADER)

Build a deep learning model to recognize and change hair color.

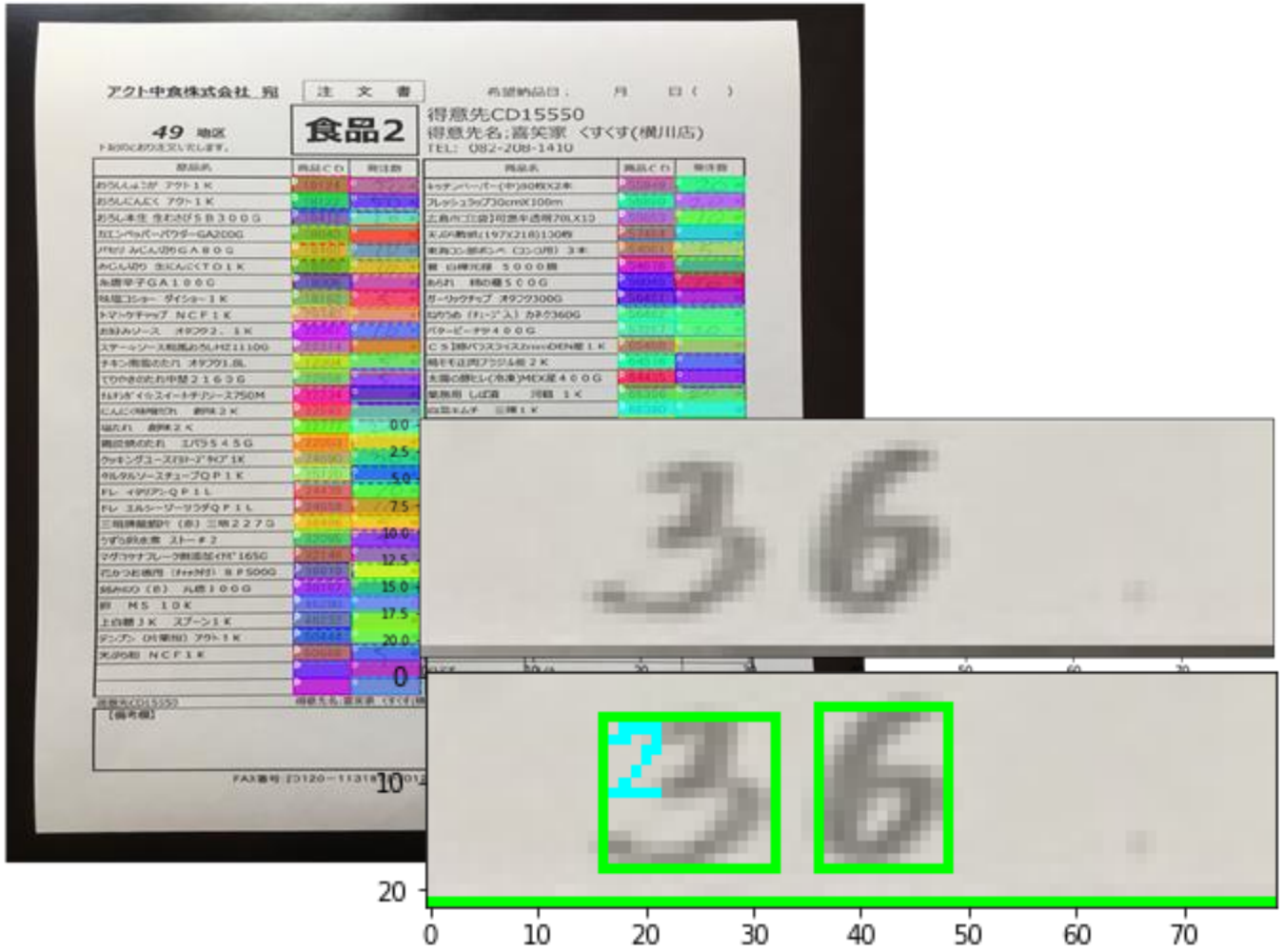
1. * Creating a deep learning model: train with more than 1 million annotated images
2. * Develop a mobile app: (convert deep learning model to coreml)

Video: https://drive.google.com/open?id=1zEbqnFHdOjx2dYE179wf6JrF0cGRimcy

4. FORM DETECTION
(Deep Learning, Python, iOS | PROJECT LEADER)

There are 2 main parts including a deep learning model of digital and handwriting number (Tesseract), an iOS app.

- * Using Mark-RCNN in training phrase to detecting boxes of number
- * MNIST and Tesseract.
- Source: https://github.com/quocnh/OCR_DeepLearning/blob/master/ocr_note.ipynb



5. CAR BUTTONS INSPECTION – TOYOTA JAPAN (Deep Learning, Python | PROJECT MEMBER)

This project aims to create a deep learning model to recognize tiny errors on the surface of different type of buttons.

図1: A-OFF button

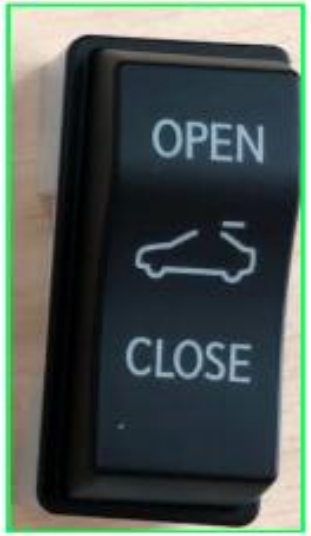


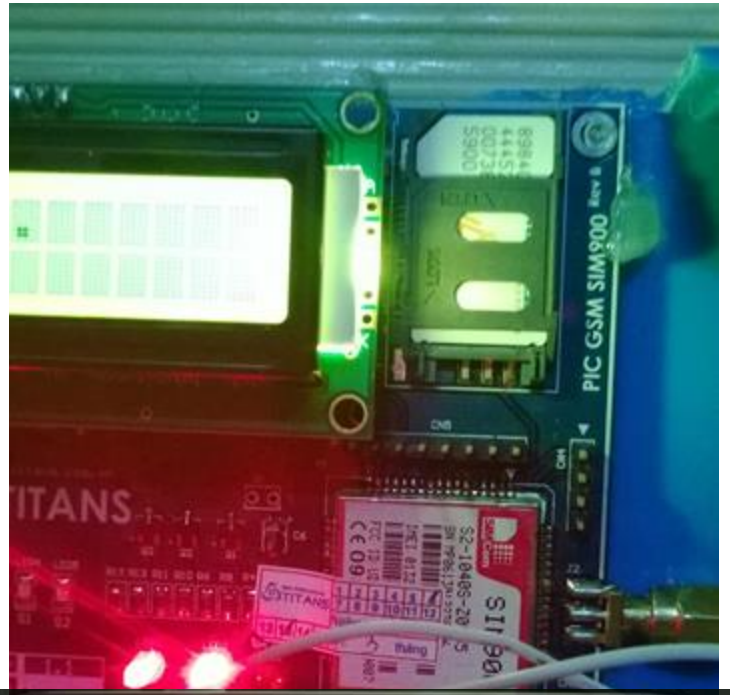
図2: O/C button

図3: A-OFF button error

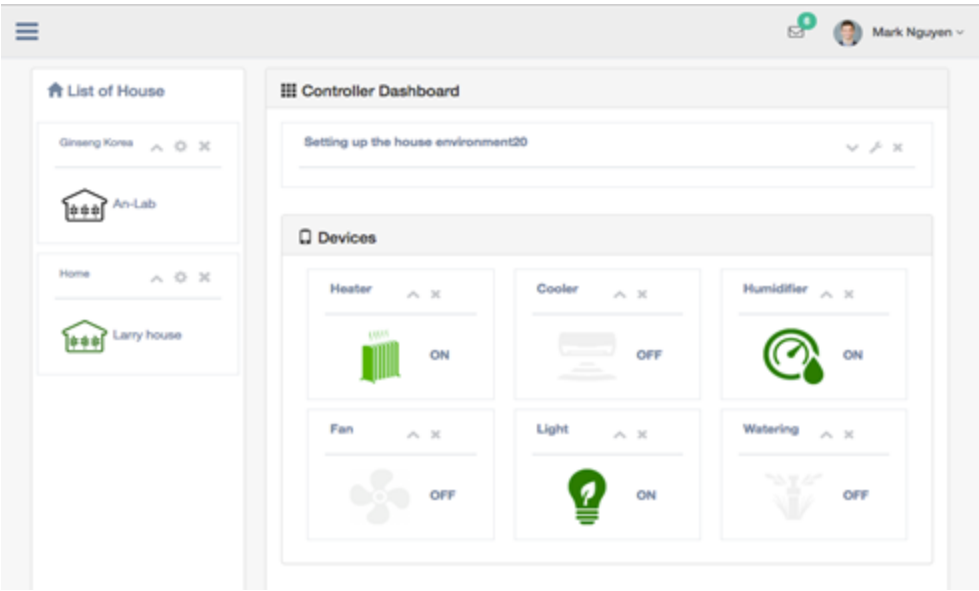
図4: O/C button error

6. SMART HOME SYSTEM WITH ARM MICROCONTROLLER (Android, C++, IoT)

Building a smart home system based on ARM Controller embedded system; automatically turn on/off house light, detecting fire, temperature, human, thief… using PIR sensor; send and receive SMS message using GSM module; develop Android application to control all devices in home from anywhere using Internet connection.
Source:
https://github.com/quocnh/smarthome-ver-tablet

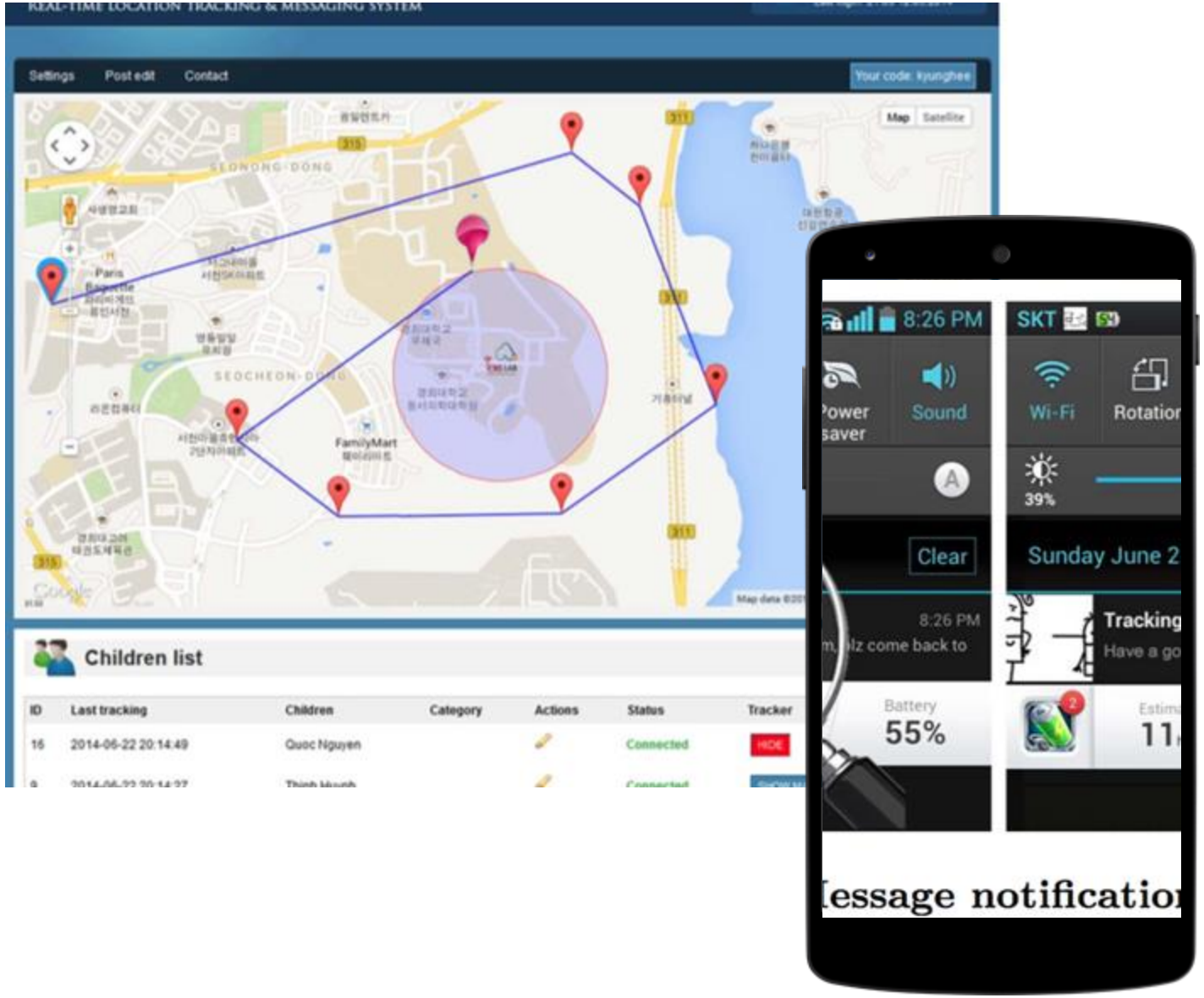


7. GREEN HOUSE
(Android, C++, Python, PhP, Bootstrap, IoT | SELF PROJECT)



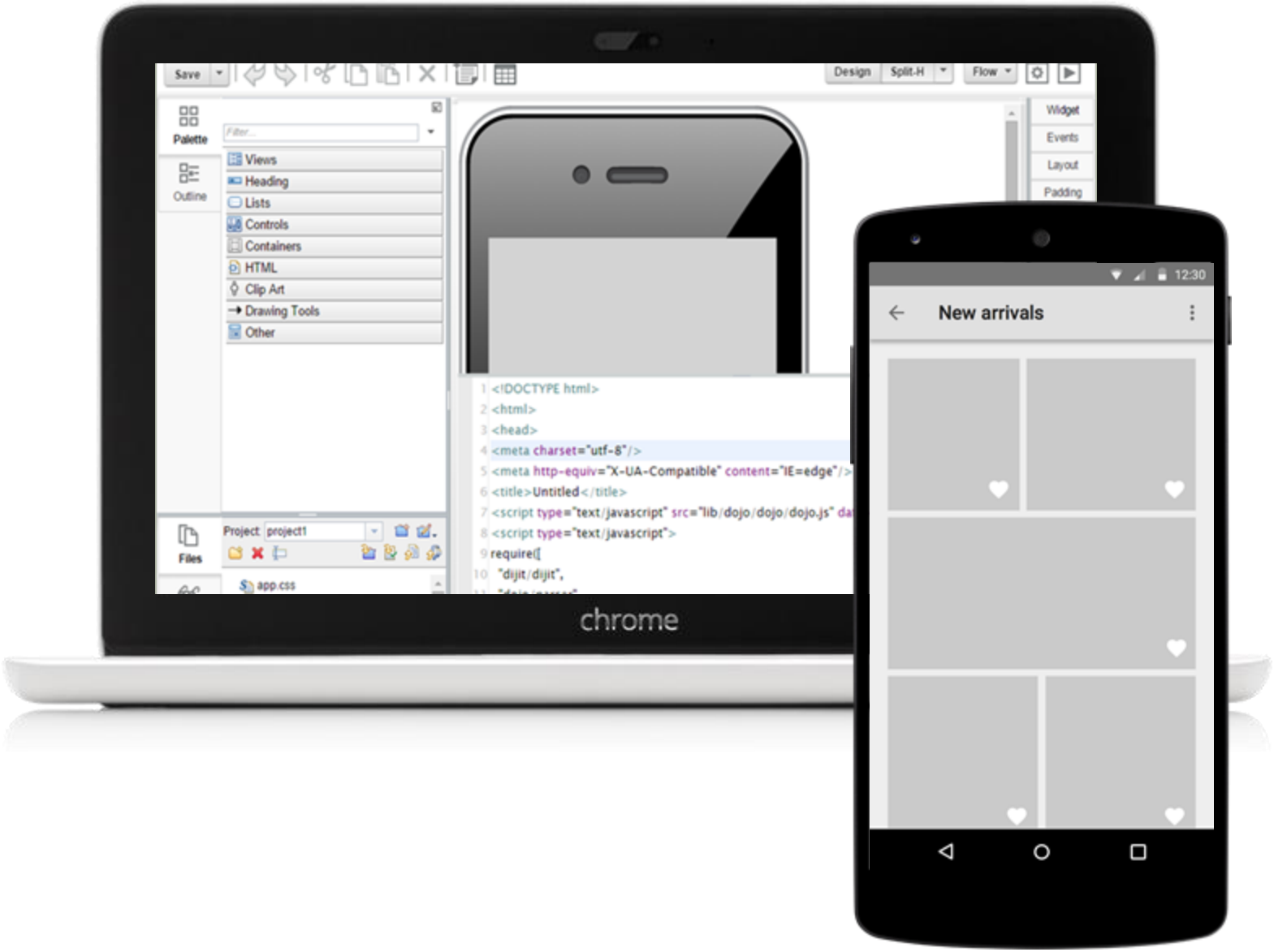
8. CHILDREN TRACKING (Android, PHP) | PROJECT LEADER

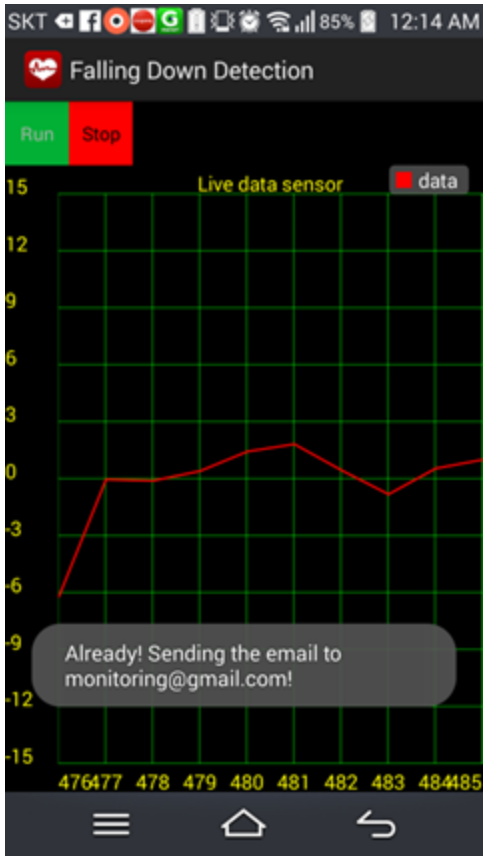
Present a cloud-based real-time location tracking and messaging system (CRLTMS), that consists a cloud-based push messaging service, namely Google Cloud Messaging (GCM), web server, database and GPS navigation data. An Android-based child-care application case study shows that the effectiveness of the system, which enables the server tracks and communicates synchronously to smart-phones in real-time manner.
Source:
https://github.com/quocnh/child-tracking

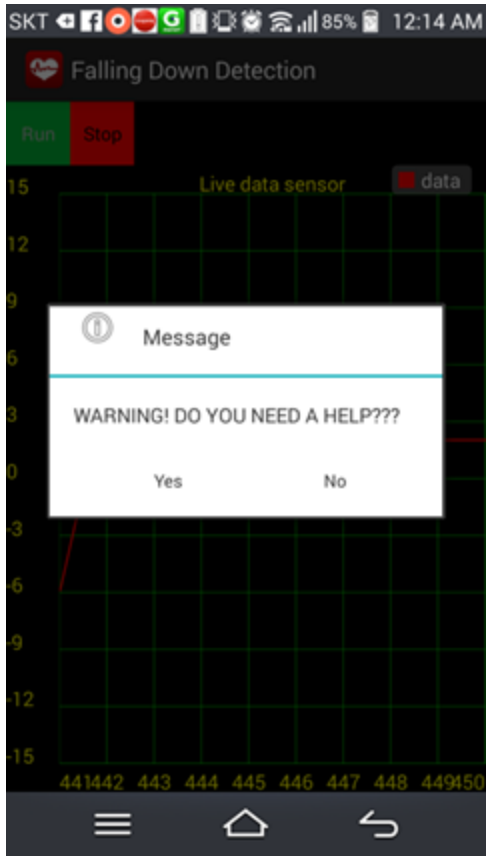


9. REAL-TIME MOBILE CLOUD SERVICE PLATFORM (Android, C++, JavaScript, HTML)

Recently, mobile services have been increasing rapidly. Mobile devices and applications are built in different platforms (e.g. Android, IOS or Windows Phone, etc.) which required advanced knowledge for developing applications. To overcome this issue, we introduce a framework that supports users to easily adapt to various platforms without having advanced knowledge. In addition, we integrate a Push Message Service on the cloud into this framework, not only in order to expand the variety of services to cloud users but also provide more service types for user demands. Our framework consists of both user interface design and application deployment across multiple mobile platforms based on Apache Cordova platform.







https://github.com/icqrx/FallingDownDetection_SoftwareEngineering

Falling Down (Android, PHP)

The purpose of fall detection mechanism uses the sensors available in a typical smartphone to detect whether an accident has occurred.

https://github.com/icqrx/TMS

Transportation Mobile System

This project helps deliver get the list of order and delivery to the customer quickly.